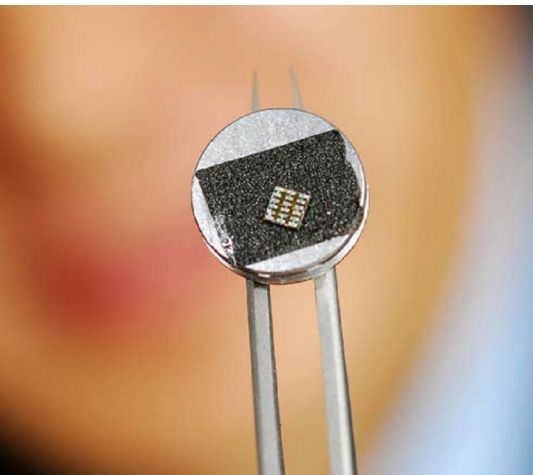


# NANO/MICRO GENERATORS FOR MEDICAL IMPLANTS

**Our body produces energy in various forms. Just a small fraction of this energy is sufficient to power implanted medical devices like pacemakers. Take a closer look at the tiny power plants harnessing the energy of our body**



*Close-up image shows a prototype direct-current nanogenerator fabricated by Georgia Tech researchers using an array of zinc oxide nanowires (Credit: Gary Meek, Georgia Tech)*

## ■ RATHINDRA NATH BISWAS

**N**anotechnology has huge potential. But nanosize machines will need a power source that is better than batteries and measures just billionths of a metre. The good news is that the waste energy, produced by our body in the form of vibrations or even the human pulse, could provide sufficient power to run such tiny gadgets. Arrays of piezo-electric nanowires could capture and

transmit that waste energy to nanodevices. These power plants are called 'nanogenerators.'

The idea of harvesting the energy produced by our body came about 90 years ago. The watchmaker in the 1920s who devised the self-winding wristwatches, was much ahead of time. His idea led to harvesting of the mechanical energy from the wearer's moving arm and putting it to work in rewinding the watch spring.

The human body is a source of power in the form of mechanical energy, heat energy, vibration energy, chemical energy (in the form of glucose) and hydraulic energy of the circulatory system. Just a small fraction of this energy when converted into electricity, is sufficient to power many types of nanodevices, as nanodevices consume very little power in the range of nanowatts to microwatts.

## Applications of nanogenerators

Medical devices are likely to be a major application for nanogenerators. People fitted with battery-powered pacemakers need surgery every few years just to get the battery replaced. That can cost as much as Rs 80,000 a time. Also, the patients have to go through the pain of having their chest cavity opened to get the work done. By adding a few sensors, the body's inner power can be harnessed to convert it into electricity—in the range of tens of microwatts to a few milliwatts—and recharge the pacemaker's battery so it doesn't need replacement. Implantable wireless nanosensors could continuously monitor a patient's blood glucose level for diabetics.

Other than body movement, various routes are being tried by researchers for power generation on mini scale. Some of these are random vibrations or motions near a roadway, temperature gradient (ground temperature several metres below the surface is fairly constant whereas ambient temperature varies), biochemistry and external energy sources such as ultrasonic waves